

End-to-End Stochastic Predict-Then-Optimize for Cost-Efficient Water-Energy Resource Scheduling

Shunyu Wu^{1b}, Jingcheng Wang^{1b}, Haotian Xu^{1b}, Yanjiu Zhong, and Jun Rao^{1b}

Abstract—Compared to other energy-intensive industries, urban water supply systems are unique due to the strong temporal coupling with the power grid, which intensifies grid stress during peak hours. This paper addresses the water-energy nexus by developing a water-energy resource scheduling model for whole-process water supply that incorporates electricity cost prediction uncertainties. Meanwhile, to minimize the scheduling cost in an end-to-end approach, we propose the Stochastic Predict-then-Optimize (SPTO), which aligns prediction distribution with the optimality gap by integrating probabilistic forecasting with stochastic optimization. Unlike traditional two-stage approaches, SPTO establishes a closed-loop feedback between forecasting and optimization, reducing cumulative errors and improving real-time adaptability. To enable end-to-end training, we derive the SPTO+, a differentiable convex upper-bound surrogate function based on Lagrangian duality, which provides surrogate gradient computation for end-to-end training. Experiments on real-world systems demonstrate that SPTO significantly reduces operational cost and decision regret, effectively coordinating water operations with dynamic grid conditions to support cost-efficient and grid-responsive scheduling.

Index Terms—Predict-then-optimize, urban water-energy system, grid load, cost-efficient scheduling.

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NOMENCLATURE

Indices

i	Index for reservoirs, $i = A, B, C$
J	Set of all pumps
j	Index for pumps, $j = 1, 2, \dots, 18$
J_d	Set of distribution pumps, $J_d = \{9, 10, \dots, 18\}$
$J_{d,A}$	Set of pumps for Water Plant A
$J_{d,B}$	Set of pumps for Pump Station B
$J_{d,C}$	Set of pumps for Pump Station C
K	Set of filtering tanks, $K = \{s, v, a\}$
k	Index for filtering tanks, $k = s, v, a$
N	Set of distribution network nodes
n	Index for distribution nodes, $n \in N$
T	Set of time steps
t	Time step index, $t = 1, 2, \dots, T$

Parameters

$\bar{\phi}_j, \bar{q}_n$	Reference flow rates for pumps and nodes
ΔI_k^t	Change in water level in tank k at time t
δ	Maximum allowed deviation from initial water level
δ^-, δ^+	Allowable shortage and excess margins
η	Evaporation rate in tanks
$\hat{c}^t$	Predicted electricity price at time t
τ_k	Time lag for outflow of tank k
a_k, b_k	Head loss coefficients for tank k
c_n	Coefficient for head loss at node n
D^t	Total demand at time t
$H_{\text{rated},j}$	Rated head of pump j at zero flow
k_j	Head loss coefficient for pump j
$L_{r_i}^{\text{low}}, L_{r_i}^{\text{high}}$	Minimum and maximum water levels for reservoir i
P_j	Power consumption of pump j
Q_j	Maximum flow capacity of pump j
Q_k^L, Q_k^H	Lower and upper flow limits for tank k
S_i	Storage capacity of reservoir i
W_k^{ini}	Initial water level in tank k

Variables

ΔI_{r_i}	Change in water level of reservoir i
Δt_j^t	Duration of pump j 's operation at time t
ϕ_j^t	Flow rate of pump j at time t
H_k^t	Head at tank k at time t
H_n^t	Head at distribution node n at time t
$q_{in,k}^t$	Inflow to tank k at time t

$q_{out,k}^t$	Outflow from tank k at time t
u_j^t	Binary decision variable indicating whether pump j is active at time t
w_k^t	Water level in tank k at time t

I. INTRODUCTION

MANAGING peak demand is a critical concern in modern smart grid operations [1], [2]. As the electrification of infrastructure grows, power grids must increasingly coordinate with large, flexible, and often uncertain energy loads [3]. Among these, the urban water supply system emerges as a significant and controllable electricity consumer [4], [5]. According to national estimates, water-related infrastructure consumes approximately 7–8% of total electricity in the United States [6]. In specific urban regions, water pumping alone can account for up to 30% of local utility energy consumption [7], providing potential for demand-side flexibility. More importantly, this load is highly time-concentrated. Studies report that during peak electricity periods—typically aligned with high-price hours—the contribution of water systems to total grid load can reach 15–20% [8].

This temporal coupling intensifies core operational challenges in smart grids, such as maintaining voltage stability, managing frequency deviations, and ensuring sufficient ramping capacity, especially under high renewable penetration [9]. Without coordination, this leads to greater reliance on expensive peaking generators and higher system risk during renewable fluctuations [10], [11]. Therefore, coordinated scheduling of water systems is essential to provide flexible load shifting that supports grid stability, with the extra benefit of reducing electricity costs. However, such coordination must respect hydraulic constraints and process delays within the water system, making joint optimization both valuable and challenging [12].

While co-optimization models aim to shift pumping away from expensive periods, most of them assume full knowledge of future prices, which is a strong assumption that contradicts the volatility and price unpredictability of real-world power markets [13], [14], [15]. This gap undermines their practical effectiveness in real-world smart grids, where electricity prices are influenced by stochastic factors such as renewable variability, demand spikes, and market bidding behavior [16], [17].

One standard approach to handle such uncertainty is the two-stage predict and optimize framework [16], [18], [19]. It uses a forecasting model to predict electricity prices and then solves an optimization problem with those predicted values. This modular design is easy to implement, but suffers from a fundamental misalignment. Traditional prediction losses, such as mean absolute error (MAE) or negative log-likelihood (NLL), measure forecast accuracy in isolation and do not reflect the impact of forecast errors on scheduling quality.

However, in grid-interactive infrastructure systems, the effectiveness of forecasts is not only judged by statistical metrics, but also by their contribution to operational decisions [20]. This has led to the emergence of Predict-then-Optimize (PTO) methods [21], which combine prediction with optimization through regret-based training losses. The

PTO ensures that the forecasting model improves outcomes that matter directly to the optimization problem. However, existing PTO methods face two key limitations when applied to realistic energy systems.

First, the regret function is often non-differentiable with respect to the predicted inputs [22], [23], especially in problems with discrete decisions or piecewise objectives. This prevents direct gradient computation and requires the use of surrogate loss functions or differentiable optimization layers [24], [25], [26], [27]. Second, and more critically, most PTO approaches rely on point predictions of uncertain parameters. That is, they train the forecasting model to output a single best-estimate price and then optimize accordingly. This ignores the distributional uncertainty inherent in electricity price forecasts, which can be large and asymmetric, especially during price spikes or load changes. Inability to model distributional uncertainty, leading to high rates of regret, especially during rare but critical events such as periods of grid stress or market volatility [28], [29].

This paper presents a decision-aware optimization framework for water-energy scheduling under cost uncertainty in smart grids. Urban water systems represent a major controllable load, and their pumping operations often coincide with periods of high electricity demand. To reduce peak grid stress, this work extends the Predict-Then-Optimize (PTO) framework to a stochastic setting, where uncertain electricity prices are modeled as probability distributions. The main contributions are summarized as follows:

- 1) A whole-process water-energy scheduling model is developed that satisfies hydraulic constraints and leverages electricity price responsiveness for demand-side flexibility in smart grids. This formulation enables pump scheduling to adapt to electricity price variations, contributing to peak load reduction and demand-side flexibility in the smart grid.
- 2) A learning-based optimization framework is introduced, where electricity prices are predicted as probability distributions rather than point estimates. These forecasts are directly embedded into the optimization problem through a decision-focused loss. The proposed framework improves the alignment between prediction and scheduling under uncertain market conditions, enhancing the robustness of demand-side operations.
- 3) To address the non-differentiability of regret-based objectives, a smooth surrogate loss is derived. This formulation enables gradient-based training of the forecasting model using decision-aware signals. The surrogate maintains fidelity to the original regret objective while supporting efficient model optimization.
- 4) The proposed framework is evaluated under a real-world water-energy scheduling scenario. Results show that the proposed method achieves lower decision regret and improved cost performance compared to deterministic and two-stage baselines. These findings support the integration of data-driven decision models into grid-interactive infrastructure systems.

Section II describes the problem formulation. Section III presents the whole-process water-energy scheduling model. Section IV introduces the proposed learning-based Stochastic

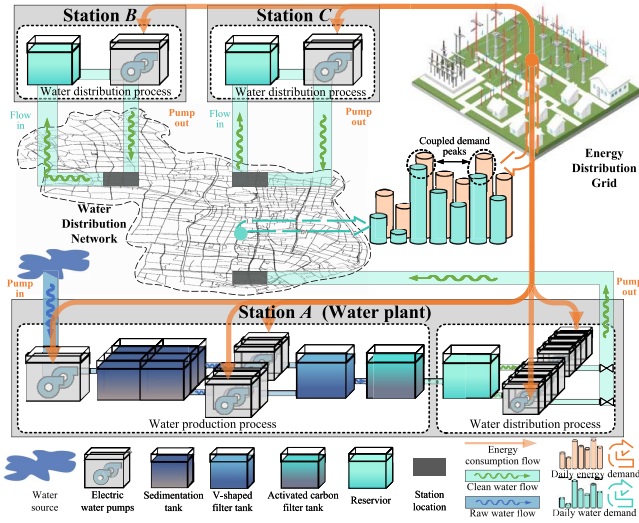


Fig. 1. Overview of the real-world study area $\mathcal{H}$. Station A handles the raw water treatment through sequential stages and the clean water distribution from the reservoir. Stations B and C just focus on water distribution to end users. All stations operate multiple pumps for significant electricity consumption. The energy demand from pumping operations exhibits temporal coupling with daily water usage patterns, leading to coinciding peak loads in both water and energy systems. This coupling motivates the proposed cost-efficient scheduling model.

Predict-then-optimizer (SPTO) framework, including the probabilistic forecasting model, cost-aware optimization, and the SPTO+ loss function. Section V provides experimental results. Section VI concludes the paper.

II. PROBLEM FORMULATION

This study aims to address the coupled peak demand problem in water systems and electric power grids. Urban water systems consume significant energy for pumping and distribution, which can intensify power grid loads during peak electricity demand periods. The study area is a specific urban water supply district $\mathcal{H}$ in a mega Asian city, covering approximately 20 km² and serving around 500,000 residents. Its water distribution network (WDN) includes thousands of nodes and pipes, along with a set of essential production and delivery facilities, such as pumping stations, reservoirs, and treatment tanks. Electricity costs for operating these water pumps account for a significant part of the system's total operational expenses.

To moderate peak-period loads and reduce operational costs, we formulate a cost-aware water-energy scheduling problem as a parametric stochastic optimization model. The objective is to minimize the expected operational cost:

$$\begin{aligned} \min_u \quad & \mathbb{E}_{c \sim Q}[c^\top u], \\ \text{s.t.} \quad & u \in \tilde{\mathcal{U}}, \end{aligned} \quad (1)$$

where u represents the decision variables (pump operations), and c denotes cost parameters (electricity prices) to be predicted for optimization. The set $\tilde{\mathcal{U}}$ defines operational constraints, including pump limits, water balance requirements, and pressure loss constraints.

In practice, the true parameter c is unknown and needs to be predicted with a forecasting model. The common approach

for most optimization problems with unknown parameters is the *two-stage* method, which predicts unknown parameters and incorporates them into the optimization model. The problems are then solved through various algorithms or solvers. Consider $\hat{c}$ as the predicted parameter, the predicted optimal decisions are made by solving $u^*(\hat{c}) = \arg \min_{u \in \tilde{\mathcal{U}}} \hat{c}^\top u$.

However, the existence of the $\arg \min$ operator in $u^*(\hat{c})$ means that even small errors in the predicted parameter cost vector $\hat{c}$ can significantly affect the optimal decision. Unlike smooth mappings, the solution to an optimization problem is often the piecewise constant function for the cost coefficients, dividing the decision space into distinct regions. Small variations in $\hat{c}$ may push it across a region boundary, causing a sudden change in the optimal decision and resulting in a notable optimality gap.

To evaluate the impact of such an optimality gap in an *end-to-end* approach, we define the decision regret function:

$$R(c, \hat{c}) = c^\top u^*(\hat{c}) - c^\top u^*(c), \quad (2)$$

where $u^*(c) = \arg \min_{u \in \tilde{\mathcal{U}}} c^\top u$ represents the true optimal decision under the actual parameter c . Such approach is also known as the *Predict-then-optimize (PTO)*. Through the explicit definition of equation (2), the PTO approach ensures that predictions are optimized to minimize the impact of parameter uncertainty on the objective value.

The key challenge in this study, as stated in the introduction, is to acquire the surrogate gradient for the *non-differentiability* of $R(\cdot)$. Small changes in $\hat{c}$ can lead to sudden changes in $u^*(\hat{c})$. Therefore, it is essential to construct a differentiable surrogate function for acquiring the surrogate gradient, enabling end-to-end training.

III. INTEGRATED WATER-ENERGY SCHEDULING MODEL

This section presents the whole-process model of the urban water-energy co-optimization system, emphasizing the integration of energy-aware pumping decisions and hydraulically feasible water production and distribution processes. The model consists of two coupled subsystems: the *energy side*, which focuses on electricity cost-aware scheduling of pumps; and the *water side*, which captures the flow balance, pressure dynamics, and process delays across the entire treatment and supply pipeline.

A. Energy-Side: Energy and Price Costs Minimization

In water supply systems, even a single suburban district typically operates dozens of pumps in the whole process supply loop, including raw-water lifts, treatment boosters, and distribution units. At peak, these drives draw MWs of electrical power, making the district a non-trivial load in local demand-side management. Moreover, their operations often align with peak electricity demand periods, which further amplifies grid stress.

To address the impact of uncertain and time-varying electricity prices, we design the objective function to capture both the operational energy usage and its temporal coupling price cost. The optimization seeks to minimize the expected total cost across the planning horizon:

$$\min_{u \in \bar{U}} \sum_{t \in T} \sum_{j \in J} \mathbb{E}[\hat{c}^t P_j u_j^t], \quad (3)$$

where $\hat{c}^t$ denotes the stochastic electricity price at time t , modeled as a random variable with known distribution or empirical samples. P_j is the rated power of pump j , and $u_j^t \in \{0, 1\}$ is the on/off control signal indicating its operational status.

The expectation operator captures the average cost under the price uncertainty, and in practice, is approximated via a sample average approximation (SAA) approach:

$$\mathbb{E}[\cdot] \approx \frac{1}{|\Omega|} \sum_{t \in T} \sum_{j \in J} \sum_{\omega \in \Omega} (\hat{c}^{t,\omega} P_j u_j^{t,\omega}),$$

where Ω denotes a finite set of sampled price scenarios. The uncertainty of cost is predicted by probabilistic neural forecasting in section IV, which enables the capture of complex, nonlinear temporal dependencies in electricity prices. Consequently, the objective function inherently encodes inter-temporal coupling in the cost profile, which significantly increases the solution difficulty when combined with binary control decisions and physical system constraints.

This formulation is risk-neutral, but it can be naturally extended to risk-aware variants by substituting the expectation with an appropriate risk measure. Moreover, the structure is compatible with decomposition or approximation techniques such as stochastic dual dynamic programming (SDDP), depending on the underlying system dynamics and constraint structure.

The energy cost component is modeled based on pump operations constrained by mechanical and hydraulic behavior. For the fixed-frequency pumps, the flow rate ϕ_j^t for pump j at time t is limited by:

$$\phi_j^t \leq u_j^t \cdot Q_j, \quad (4)$$

where ϕ_j^t is the actual delivered flow and Q_j is the pump's maximum capacity.

Considering the practical parallel pump systems, it is essential to maintain the same head for all pumps. The head of each pump depends on the flow rate, and this relationship is initially modeled by a quadratic equation. Specifically, at time t and for each pump $j \in P$, we have:

$$H_{\text{rated},j} - k_j (\phi_j^t)^2 = H^t, \quad (5)$$

where $H_{\text{rated},j}$ is the rated head at zero flow, k_j is a constant describing how the head decreases with increasing flow for pump j , and H^t is the common head at time t . This ensures that the head for all pumps is equal at each time t . Moreover, from the equal-head condition, the following must hold:

$$k_1 (\phi_1^t)^2 = k_2 (\phi_2^t)^2 = \dots = k_m (\phi_m^t)^2, \quad (6)$$

where m denotes the number of parallel pumps. Considering that these quadratic equality constraints can be difficult to handle in the solver. Therefore, we consider a linear approximation around a chosen reference flow rate $\bar{\phi}_j$ for each pump. Using a first-order Taylor expansion of the quadratic term $(\phi_j^t)^2$ around $\bar{\phi}_j$, we approximate:

$$(\phi_j^t)^2 \approx 2\bar{\phi}_j \phi_j^t - (\bar{\phi}_j)^2.$$

Substituting this into the head equation yields a linear approximation:

$$H_{\text{rated},j} - k_j [2\bar{\phi}_j \phi_j^t - (\bar{\phi}_j)^2] \approx H^t. \quad (7)$$

Similarly, the equal-head conditions can be approximated by replacing each $(\phi_j^t)^2$ term with its linearized form. By choosing appropriate reference flow values, these linear approximations enable standard linear (or mixed-integer linear) optimization solvers to be applied. While this approximation is only exact at the chosen reference point, it provides a computationally tractable approach and can be iteratively refined if necessary.

B. Water-Side: Whole-Process Water Balance

The water side of the system is divided into two processes: distribution and production. For the water distribution subsystem, it maintains a real-time supply—demand balance to guarantee that end-user consumption is met at adequate pressure. By capturing time-lagged flow transfer and treatment-unit pressure losses, the water production subsystem aims to keep downstream reservoir levels within safe bands. Modeling of the water side focuses on the water constraints part of the optimization model.

1) *Water Distribution Process Constraints:* Consider the total demand at each time t balanced by the aggregate output of the distribution pumps:

$$D^t - \delta^- \leq \sum_{j \in J_d} \phi_j^t \leq D^t + \delta^+, \quad (8)$$

where $J_d \subset J$ indicates the subset of pumps for delivery, and δ^-, δ^+ represent the tolerated mismatch. To regulate mechanical wear, we introduce operational time variables Δt_j^t subject to:

$$\Delta t_j^t \leq M \cdot u_j^t, \quad (9)$$

$$T_{\min} \leq \sum_{t \in T} u_j^t \cdot \Delta t_j^t \leq T_{\max}. \quad (10)$$

Reservoir operations ensure safe water-level management at stations r_i , where $i \in \{A, B, C\}$. The water-level changes are calculated by inflows from pump sets and specific outflows, as follows:

$$\Delta l_{r_i}^t = \frac{\sum_{j \in J_{d,i}} \phi_j^t - q_{\text{out},i}^t}{S_i}, \quad i \in \{A, B, C\}. \quad (11)$$

Here, $J_{d,i}$ denotes pump indices associated with each reservoir, and $q_{\text{out},A}^t$ specifically represents the activated carbon filter outflow entering Reservoir A. S_i normalizes these flow differences into water-level changes. The cumulative water levels must remain within safe operational limits:

$$L_{r_i}^{\text{low}} \leq L_{r_i} - \sum_{t \in T} \Delta l_{r_i}^t \leq L_{r_i}^{\text{high}}, \quad i \in \{A, B, C\}. \quad (12)$$

At the distribution nodes $n \in N$, head loss follows a simplified Darcy-Weisbach quadratic form:

$$H_n^t \approx H^t - c_n [2\bar{q}_n q_n^t - (\bar{q}_n)^2], \quad (13)$$

where H^t is the common head from upstream pumps, and c_n captures pipeline head-loss characteristics. To ensure network operability, each node's head is constrained:

$$H_n^t \geq H_n^{\min}, \quad n \in N, \quad t \in T. \quad (14)$$

2) *Water Production Process Constraints*: The water production process involves multiple treatment stages, including sedimentation, filtration, and activated carbon absorption. All of these stages involve modeling of dynamics, time delays, and pressure requirements.

The inflow into each treatment tank $k \in \{s, v, a\}$ is determined by the cumulative flow of the associated pumps:

$$q_{in,k}^t = \sum_{j \in J_k} \phi_j^t, \quad (15)$$

where J_k denotes the set of pumps assigned to tank k . To stabilize operations, inflow bounds are enforced:

$$Q_k^L \leq q_{in,k}^t \leq Q_k^H. \quad (16)$$

The delay property of each treatment process can be further formulated through lagged outflows:

$$q_{out,k}^t = q_{in,k}^{t-\tau_k}, \quad (17)$$

where τ_k is the delay associated with sedimentation or filtration dynamics. The corresponding water-level change in tank k is given by:

$$\Delta h_k^t = \frac{q_{in,k}^t - q_{out,k}^t}{A_k}, \quad (18)$$

where A_k represents the surface area of tank k .

To prevent large fluctuations of water level in tanks, we further adopt a recursive balance relation as:

$$w_k^t = (1 - \gamma)w_k^{t-1} + q_{in,k}^t - q_{out,k}^t, \quad (19)$$

subject to stability bounds:

$$|w_k^t - W_k^{\text{ini}}| \leq \delta, \quad (20)$$

where γ is the evaporation rate, W_k^{ini} is the initial target level, and δ denotes the maximum permissible deviation.

Finally, to limit the pressure loss in each treatment tank $k \in K = \{s, v, a\}$, the Forchheimer-type equation is applied to account for the head loss as a function of flow. The outlet head at time $t \in T$ is given by:

$$H_{out,k}^t = H_{in,k}^t - \left(a_k q_{out,k}^t + b_k (q_{out,k}^t)^2 \right), \quad \forall k \in K, \quad t \in T, \quad (21)$$

where $H_{in,k}^t$ and $H_{out,k}^t$ denote the inlet and outlet heads, respectively, and $q_{out,k}^t$ is the outflow rate. Parameters a_k and b_k are the linear and quadratic head loss coefficients.

To simplify the optimization model, a first-order Taylor approximation is applied around a nominal flow value $\bar{q}_{out,k}$, resulting in:

$$H_{out,k}^t \approx H_{in,k}^t - \left[a_k q_{out,k}^t + b_k \left(2\bar{q}_{out,k} q_{out,k}^t - (\bar{q}_{out,k})^2 \right) \right], \quad (22)$$

The inlet head $H_{in,k}^t$ varies depending on the tank type: for $k = s$, it comes from the initial pumps; for $k = v$, it is provided

by the intermediate pump group; and for $k = a$, it corresponds to the outlet head of tank v . To ensure proper functioning, a constraint is applied that the outlet head of each tank must be above the required minimum level at all times:

$$H_k^t \geq H_k^{\min}, \quad k \in K, \quad t \in T \quad (23)$$

where $H_k^{\min}$ is the minimum required outlet head for tank k .

IV. END-TO-END STOCHASTIC PREDICT-THEN-OPTIMIZE FRAMEWORK

Section III established a detailed integrated model including both energy-side scheduling and water-side hydraulic operations. To address the problem of decision making under uncertainty, we propose the end-to-end Stochastic Predict-then-Optimize (SPTO) framework. As shown in Fig. 2, the framework integrates a probabilistic forecasting neural network for estimating uncertain parameters and the stage for cost-efficient scheduling optimization.

In the following sections, we first describe the optimization and prediction stages separately. After, we explain specifically how to connect the two stages by constructing a surrogate function with surrogate gradients in Section IV-C.

A. Cost-Efficient Scheduling Optimization

The scheduling model (3) involves uncertain electricity prices $\hat{c}^t$, aiming to minimize the expected total operational cost induced by pump electricity usage. Mathematically, this objective is formulated as:

$$\begin{aligned} \min_{u \in \tilde{U}} \quad & \sum_{t \in T} \sum_{j \in J} \mathbb{E}[\hat{c}^t P_j u_j^t] \\ \text{s.t.} \quad & \text{Equations (4)–(24)}, \end{aligned} \quad (24)$$

where $\hat{c}^t$ is the predicted electricity price at time t , P_j denotes the rated power of pump j , and $u_j^t \in \{0, 1\}$ denotes binary activation signals. The feasible region $\tilde{U}$ captures the complete set of constraints from Section III, including pump dynamics, reservoir and pipeline hydraulics, and operational requirements.

Note that While the Predict-then-optimize framework is originally formulated under the assumption of a convex and closed feasible region, Elmachtoub and Grigas [21] have shown that this assumption can be relaxed without loss of generality. For mixed-integer optimization problems, the origin non-convex feasible region $\tilde{U}$ can be replaced by its closed convex hull $\mathcal{U}$. Since linear optimization over $\tilde{U}$ is equivalent to linear optimization over $\mathcal{U}$, the convex surrogate loss functions can be directly applied to these problems. This equivalence enables the Predict-then-optimize framework to handle a broad range of decision-making tasks involving binary variables.

In practice, directly solving (24) requires accurate probabilistic forecasting of electricity prices, therefore motivating the following forecasting mechanism.

B. Probabilistic Forecasting of Electricity Prices

As stated in the problem formulation, the electricity price $\hat{c}$ needs to be predicted for optimization. Here, we develop a

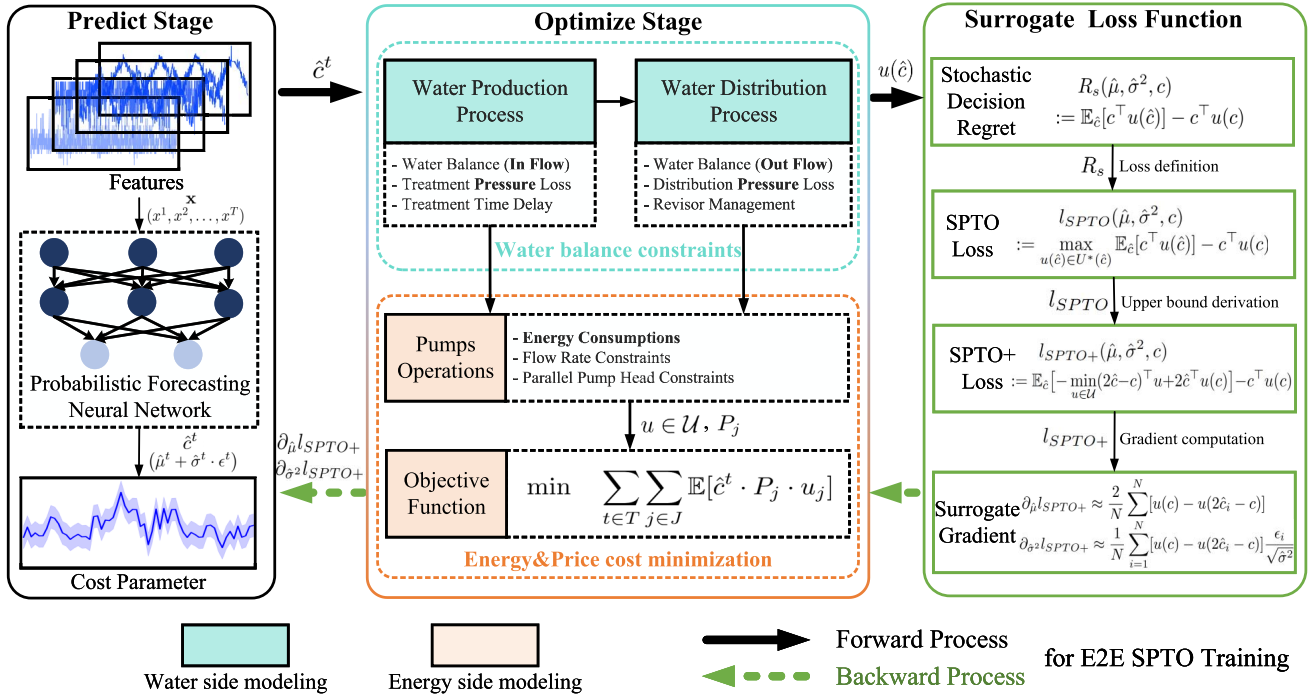


Fig. 2. Overview of the proposed Stochastic Predict-then-optimize (SPTO) framework for cost-efficient water-energy scheduling. The framework integrates a probabilistic forecasting model, an optimization model with detailed water–energy constraints, and a surrogate regret-based loss to enable end-to-end training for cost-efficient scheduling.

probabilistic forecasting model to produce output as probability distributions rather than a single point value. Unlike the normal forecasting model, probabilistic forecasting captures the uncertainty of the predictions, which is valuable for handling the highly uncertain nature of electricity prices.

In this paper, we formulate the prediction model as $f(\Theta)$ for electricity prices forecasting. For the primary model, we utilize the Bidirectional Long Short-Term Memory (BiLSTM) network, which has been proven effective for the sequential nature of electricity price in existing studies [30], [31], [32]. Though utilization may come with higher computational complexity, the accuracy improvements through BiLSTM are important for capturing the dynamics of electricity price data.

$$h^t = \text{BiLSTM}(x^t, h^{t-1}; \Theta_{\text{lstm}}), \quad (25)$$

where h^t denotes the hidden state at time t , and Θ_{lstm} represents the parameters of the BiLSTM layers. To enhance the model performance for capture temporal dependencies among features, the multi-head attention module [33] is incorporated as:

$$\tilde{h}^t = \text{Attention}(h^t, h; \Theta_{\text{att}}), \quad (26)$$

where $\tilde{h}^t$ is the output of the attention layer, and Θ_{att} denotes the parameters of the attention module. Note that both BiLSTM and the attention module are commonly used in the field of deep learning, so we simplify their formulas as this is not the innovation of this paper.

The model generates the mean and variance parameters of the prediction distribution through two separate fully connected networks, incorporating dropout and LeakyReLU

activations for enhanced generalization. The parameters are defined as follows:

$$\hat{\mu}^t = \text{Linear}_{\mu}(\tilde{h}^t; \Theta_{\mu}), \quad \hat{\sigma}^t = \exp(\text{Linear}_{\sigma}(\tilde{h}^t; \Theta_{\sigma})) + 10^{-6}, \quad (27)$$

where $\text{Linear}_{\mu}(\cdot)$ and $\text{Linear}_{\sigma}(\cdot)$ are linear layers parameterized by Θ_{μ} and Θ_{σ} . The use of the exponential function ensures positivity for σ^t , and a small constant is added for numerical stability.

To enable gradient-based optimization, we utilize the reparameterization trick to express $\hat{c}^t$ as a differentiable function of the predicted mean $\hat{\mu}^t$ and standard deviation $\hat{\sigma}^t$:

$$\hat{c}^t = \hat{\mu}^t + \hat{\sigma}^t \cdot \epsilon^t, \quad \epsilon^t \sim \mathcal{N}(0, 1), \quad (28)$$

where ϵ^t is drawn from a standard normal distribution. This formulation allows the model to generate cost predictions $\hat{c}^t$ while keeping differentiability with respect to the prediction parameters $\hat{\mu}^t$ and $\hat{\sigma}^t$.

C. Mathematical Formulations of Stochastic Predict-Then-Optimize

For optimizing the cost-efficient resource scheduling model in an end-to-end approach, we give the mathematical formulations of the Stochastic Predict-Then-Optimize (SPTO) framework to derive the surrogate gradient for training. Given the predicted cost distribution $\hat{c} \sim \mathcal{N}(\hat{\mu}, \hat{\sigma}^2 I)$, we evaluate the performance of a decision-making policy using the stochastic decision regret:

$$R_s(\hat{\mu}, \hat{\sigma}^2, c) := \mathbb{E}_{\hat{c}}[c^\top u(\hat{c})] - c^\top u(c), \quad (29)$$

where $R_s(\cdot)$ denotes the stochastic version of $R(\cdot)$ in equation (2). $u(\hat{c})$ represents the decision made under the predicted cost vector $\hat{c}$, and $u(c)$ is the optimal decision under the true cost vector c . This loss quantifies the expected regret induced by prediction uncertainty.

However, a challenge arises when the optimization solution set $U^*(\hat{c})$ is not a singleton. In this case, the selection of a specific solution $u^*(\hat{c}) \in U^*(\hat{c})$ can lead to inconsistencies in the regret computation. For instance, a degenerate prediction $\hat{c} = 0$ still makes $U^*(0) = \mathcal{U}$, resulting in a trivial regret value.

To resolve this ambiguity, we define the SPTO loss using a worst-case selection over all possible optimal solutions. This unambiguous formulation ensures consistency and robustness in the regret evaluation:

$$l_{SPTO}(\hat{\mu}, \hat{\sigma}^2, c) := \max_{u(\hat{c}) \in U^*(\hat{c})} \mathbb{E}_{\hat{c}}[c^\top u] - c^\top u(c). \quad (30)$$

While l_{SPTO} provides a rigorous measure of decision regret under prediction uncertainty, it is generally non-differentiable due to the arg min operator within $u^*(\hat{c})$. To address this, we derive a convex upper bound surrogate function for l_{SPTO} . The surrogate function approximates the regret loss while preserving differentiability.

Proposition 1 (Surrogate Loss Function for l_{SPTO}): Let $\mathcal{U} \subseteq \mathbb{R}^d$ be convex and compact, and let $c \in \mathbb{R}^d$ be the true cost vector. For $\hat{c} \sim N(\hat{\mu}, \hat{\sigma}^2 I)$, consider the SPTO loss as:

$$l_{SPTO}(\hat{\mu}, \hat{\sigma}^2, c) := \max_{u(\hat{c}) \in U^*(\hat{c})} \mathbb{E}_{\hat{c}}[c^\top u(\hat{c})] - c^\top u(c),$$

where $U^*(\hat{c}) = \arg \min_{u \in \mathcal{U}} \hat{c}^\top u$. The surrogate loss is:

$$l_{SPTO+}(\hat{\mu}, \hat{\sigma}^2, c) := \mathbb{E}_{\hat{c}}[-\min_{u \in \mathcal{U}}(2\hat{c} - c)^\top u + 2\hat{c}^\top u(c)] - c^\top u(c),$$

and satisfies:

$$l_{SPTO}(\hat{\mu}, \hat{\sigma}^2, c) \leq l_{SPTO+}(\hat{\mu}, \hat{\sigma}^2, c).$$

Proof: By definition:

$$l_{SPTO}(\hat{\mu}, \hat{\sigma}^2, c) = \max_{u(\hat{c}) \in U^*(\hat{c})} \mathbb{E}_{\hat{c}}[c^\top u(\hat{c})] - c^\top u(c). \quad (31)$$

For each realization $\hat{c}$, the set $U^*(\hat{c})$ comprises all optimal solutions to $\min_{u \in \mathcal{U}} \hat{c}^\top u$. Among these optimal solutions, l_{SPTO} selects the one that maximizes $c^\top u(\hat{c})$.

Suppose $\mathcal{U} = \{u \in \mathbb{R}^d: Au \leq b\}$ for some $A \in \mathbb{R}^{m \times d}$ and $b \in \mathbb{R}^m$. The minimization problem for a given $\hat{c}$ is:

$$\min_{u \in \mathcal{U}} \hat{c}^\top u \quad \text{s.t. } Au \leq b. \quad (32)$$

Its Lagrangian function is:

$$\mathcal{L}(u, \lambda) = \hat{c}^\top u + \lambda^\top (Au - b), \quad \lambda \geq 0. \quad (33)$$

The dual function is:

$$g(\lambda) = \inf_u \mathcal{L}(u, \lambda) = \begin{cases} -\lambda^\top b & \text{if } A^\top \lambda = -\hat{c}, \\ -\infty & \text{otherwise.} \end{cases} \quad (34)$$

By weak duality:

$$\min_{u \in \mathcal{U}} \hat{c}^\top u \geq \max_{\lambda \geq 0} g(\lambda). \quad (35)$$

We have:

$$l_{SPTO}(\hat{\mu}, \hat{\sigma}^2, c) = \mathbb{E}_{\hat{c}} \left[\max_{u(\hat{c}) \in U^*(\hat{c})} c^\top u(\hat{c}) \right] - c^\top u(c). \quad (36)$$

Since $u(\hat{c})$ is the solution of $\min_{u \in \mathcal{U}} \hat{c}^\top u$, we know $\hat{c}^\top u(\hat{c}) \leq \hat{c}^\top u(c)$ for $u(c)$ defined by minimizing $c^\top u$. Consider the first-order perturbed linear form $(2\hat{c} - c)^\top u$, we claim that:

$$c^\top u(\hat{c}) - c^\top u(c) \leq -\min_{u \in \mathcal{U}} (2\hat{c} - c)^\top u + 2\hat{c}^\top u(c) - c^\top u(c). \quad (37)$$

Intuitively, it bounds the difference $c^\top u(\hat{c}) - c^\top u(c)$ by shifting the costs from $\hat{c}$ to $2\hat{c} - c$ and using the minimal value of $(2\hat{c} - c)^\top u$ over $\mathcal{U}$. Since $u(\hat{c})$ is optimal for $\hat{c}$, comparing it to a solution optimal for c and adjusting via $(2\hat{c} - c)$ yields this upper bound. Taking expectation with respect to $\hat{c}$, we obtain:

$$\mathbb{E}_{\hat{c}}[c^\top u(\hat{c}) - c^\top u(c)] \leq \mathbb{E}_{\hat{c}}[-\min_{u \in \mathcal{U}} (2\hat{c} - c)^\top u + 2\hat{c}^\top u(c)] - c^\top u(c). \quad (38)$$

Since the left-hand side represents the worst-case chosen $u(\hat{c})$ in $U^*(\hat{c})$ (i.e., l_{SPTO}), we have:

$$l_{SPTO}(\hat{\mu}, \hat{\sigma}^2, c) \leq \mathbb{E}_{\hat{c}}[-\min_{u \in \mathcal{U}} (2\hat{c} - c)^\top u + 2\hat{c}^\top u(c)] - c^\top u(c) \quad (39)$$

Therefore:

$$l_{SPTO}(\hat{\mu}, \hat{\sigma}^2, c) \leq l_{SPTO+}(\hat{\mu}, \hat{\sigma}^2, c).$$

This completes this proof. $\blacksquare$

In the following proposition, we aim to show that l_{SPTO} and l_{SPTO+} are fisher consistent. Fisher consistency means that if the model can achieve the “perfect prediction” scenario, the set of minimizers of l_{SPTO} coincides with the set of minimizers of l_{SPTO+} .

Proposition 2 (Fisher Consistency of l_{SPTO+}): Let $\mathcal{U} \subseteq \mathbb{R}^d$ be convex and compact, and let $c \in \mathbb{R}^d$ be the true cost vector. For $\hat{c} \sim N(\hat{\mu}, \hat{\sigma}^2 I)$, the l_{SPTO} and l_{SPTO+} are defined as:

$$l_{SPTO}(\hat{\mu}, \hat{\sigma}^2, c) := \max_{u \in U^*(\hat{c})} \mathbb{E}_{\hat{c}}[c^\top u] - c^\top u(c),$$

$$l_{SPTO+}(\hat{\mu}, \hat{\sigma}^2, c) := \mathbb{E}_{\hat{c}}[-\min_{u \in \mathcal{U}} (2\hat{c} - c)^\top u + 2\hat{c}^\top u(c)] - c^\top u(c).$$

If $\hat{\mu} = c$ and $\hat{\sigma}^2 = 0$ (perfect prediction), then:

$$\arg \min_{\hat{\mu}, \hat{\sigma}^2} l_{SPTO}(\hat{\mu}, \hat{\sigma}^2, c) = \arg \min_{\hat{\mu}, \hat{\sigma}^2} l_{SPTO+}(\hat{\mu}, \hat{\sigma}^2, c).$$

Proof: For l_{SPTO} , by definition:

$$l_{SPTO}(\hat{\mu}, \hat{\sigma}^2, c) = \max_{u \in U^*(\hat{c})} \mathbb{E}_{\hat{c}}[c^\top u] - c^\top u(c). \quad (40)$$

Since $u(c)$ minimizes $c^\top u$, we have for any $u \in \mathcal{U}$ and any $\hat{c}$:

$$c^\top u(\hat{c}) \geq c^\top u(c). \quad (41)$$

Thus:

$$\mathbb{E}_{\hat{c}}[c^\top u(\hat{c})] \geq c^\top u(c). \quad (42)$$

Therefore, the l_{SPTO+} is nonnegative.

Under perfect prediction, let $(\hat{\mu}^*, \hat{\sigma}^{2*})$ be $\hat{c} = c$. Then every realization of $\hat{c}$ equals c , and the set $U^*(c)$ is simply the set of optimal solutions to $c^\top u$. Since $u(c)$ is an optimal solution under true cost c , we have:

$$l_{SPTO}(c, 0, c) = \max_{u \in U^*(c)} \mathbb{E}_{\hat{c}=c} [c^\top u] - c^\top u(c). \quad (43)$$

Considering that $\hat{c} = c$, for any u in $U^*(c)$, we have:

$$c^\top u = c^\top u(c) \quad (44)$$

Hence:

$$l_{SPTO}(c, 0, c) = c^\top u(c) - c^\top u(c) = 0. \quad (45)$$

Similarly, substitute $\hat{c} = c$ into l_{SPTO+} :

$$l_{SPTO+}(c, 0, c) = \mathbb{E}_{\hat{c}=c} \left[-\min_{u \in U} (2c - c)^\top u + 2c^\top u(c) \right] - c^\top u(c). \quad (46)$$

Since $(2c - c) = c$ and $\min_u c^\top u = c^\top u(c)$:

$$-\min_u (2c - c)^\top u = -\min_u c^\top u = -c^\top u(c). \quad (47)$$

Then we have:

$$l_{SPTO+}(c, 0, c) = [-c^\top u(c) + 2c^\top u(c)] - c^\top u(c) = 0. \quad (48)$$

Therefore, under perfect prediction, both l_{SPTO} and l_{SPTO+} achieve their minimal value of 0.

Fisher consistency requires that if we can realize the true cost distribution exactly (perfect prediction), then the minimizers of l_{SPTO} and l_{SPTO+} coincide. Since both l_{SPTO} and l_{SPTO+} are nonnegative and achieve their infimum of 0 at the perfect predictor, no other predictor $(\hat{\mu}, \hat{\sigma}^2)$ can yield a value less than 0 for either loss. Hence:

$$\arg \min_{\hat{\mu}, \hat{\sigma}^2} l_{SPTO}(\hat{\mu}, \hat{\sigma}^2, c) = \arg \min_{\hat{\mu}, \hat{\sigma}^2} l_{SPTO+}(\hat{\mu}, \hat{\sigma}^2, c). \quad (49)$$

This completes the proof. ■

After establishing the Fisher consistency between l_{SPTO} and l_{SPTO+} , we turn our attention to the computation of gradients for l_{SPTO+} . In the following proposition, we derive the surrogate gradient using sub-gradient of l_{SPTO+} and the chain rule.

Proposition 3 (Gradient computation for l_{SPTO+}): Let $\mathcal{U} \subseteq \mathbb{R}^d$ be convex and compact, and let $c \in \mathbb{R}^d$ be the true cost vector. For $\hat{c} \sim N(\hat{\mu}, \hat{\sigma}^2 I)$ and $l_{SPTO+}(\hat{\mu}, \hat{\sigma}^2, c) := \mathbb{E}_{\hat{c}} [-\min_{u \in \mathcal{U}} (2\hat{c} - c)^\top u + 2\hat{c}^\top u(c)] - c^\top u(c)$, the gradient of l_{SPTO+} is computed as:

$$\partial_{\hat{\mu}} l_{SPTO+} \approx \frac{2}{M} \sum_{i=1}^M [u(c) - u(2\hat{c}_i - c)], \quad (50)$$

$$\partial_{\hat{\sigma}^2} l_{SPTO+} \approx \frac{1}{M} \sum_{i=1}^M [u(c) - u(2\hat{c}_i - c)] \frac{\epsilon_i}{\sqrt{\hat{\sigma}^2}}. \quad (51)$$

Proof: For the l_{SPTO+} with $\hat{c}$, we define:

$$f(\hat{c}) := -\min_{u \in U} (2\hat{c} - c)^\top u + 2\hat{c}^\top u(c) - c^\top u(c). \quad (52)$$

Let $g(\hat{c}) := \min_{u \in U} (2\hat{c} - c)^\top u$, the minimum is reached at $u(2\hat{c} - c) \in U$. For the function $h(v) = \min_{u \in U} v^\top u$

for $v = (2\hat{c} - c)$, it is the support function of the set $-U$. Specifically, $h(v)$ is convex and piecewise linear in v . Differentiating w.r.t. v gives a subgradient $u(v)$ where $u(v)$ is any optimal solution. Here, $v = 2\hat{c} - c$, so:

$$\partial_v h(v) \ni u(v). \quad (53)$$

Since $v = 2\hat{c} - c$, we have $\frac{\partial v}{\partial \hat{c}} = 2I$. By the chain rule:

$$\partial_{\hat{c}} g(\hat{c}) = \partial_{\hat{c}} \min_u ((2\hat{c} - c)^\top u) \ni 2u(2\hat{c} - c). \quad (54)$$

Thus:

$$\partial_{\hat{c}} (-g(\hat{c})) = -2u(2\hat{c} - c). \quad (55)$$

Also:

$$\partial_{\hat{c}} (2\hat{c}^\top u(c)) = \{2u(c)\}, \quad \partial_{\hat{c}} (-c^\top u(c)) = \{0\} \quad (56)$$

Combining these subgradients:

$$\begin{aligned} \partial_{\hat{c}} f(\hat{c}) &= \partial_{\hat{c}} [-g(\hat{c})] + \partial_{\hat{c}} (2\hat{c}^\top u(c)) + \partial_{\hat{c}} (-c^\top u(c)) \\ &\ni -2u(2\hat{c} - c) + 2u(c) \\ &= 2[u(c) - u(2\hat{c} - c)]. \end{aligned} \quad (57)$$

Since l_{SPTO+} is the expectation of $f(\hat{c})$ over $\hat{c}$, linearity of expectation ensures:

$$\partial_{\hat{c}} l_{SPTO+} \ni 2[u(c) - u(2\hat{c} - c)], \quad (58)$$

we have $\hat{c} = \hat{\mu} + \sqrt{\hat{\sigma}^2} \epsilon$. Therefore:

$$\frac{\partial \hat{c}}{\partial \hat{\mu}} = I, \quad \frac{\partial \hat{c}}{\partial \hat{\sigma}^2} = \frac{\epsilon}{2\sqrt{\hat{\sigma}^2}}. \quad (59)$$

For a fixed $\hat{c}$:

$$\partial_{\hat{\mu}} f(\hat{c}) = \partial_{\hat{c}} f(\hat{c}) \frac{\partial \hat{c}}{\partial \hat{\mu}} = 2[u(c) - u(2\hat{c} - c)], \quad (60)$$

$$\partial_{\hat{\sigma}^2} f(\hat{c}) = \partial_{\hat{c}} f(\hat{c}) \frac{\partial \hat{c}}{\partial \hat{\sigma}^2} = [u(c) - u(2\hat{c} - c)] \frac{\epsilon}{\sqrt{\hat{\sigma}^2}}. \quad (61)$$

Since l_{SPTO+} involves an expectation over $\hat{c}$, we approximate by M samples $\hat{c}_i = \hat{\mu} + \sqrt{\hat{\sigma}^2} \epsilon_i$:

$$l_{SPTO+} \approx \frac{1}{M} \sum_{i=1}^M f(\hat{c}_i). \quad (62)$$

Differentiating $\hat{\mu}$:

$$\partial_{\hat{\mu}} l_{SPTO+} \approx \frac{1}{M} \sum_{i=1}^M \partial_{\hat{\mu}} f(\hat{c}_i) = \frac{2}{M} \sum_{i=1}^M [u(c) - u(2\hat{c}_i - c)]. \quad (63)$$

Similarly, for $\hat{\sigma}^2$:

$$\partial_{\hat{\sigma}^2} l_{SPTO+} \approx \frac{1}{M} \sum_{i=1}^M \partial_{\hat{\sigma}^2} f(\hat{c}_i) = \frac{1}{M} \sum_{i=1}^M [u(c) - u(2\hat{c}_i - c)] \frac{\epsilon_i}{\sqrt{\hat{\sigma}^2}}. \quad (64)$$

This completes the proof. ■

To demonstrate the structure of the proposed SPTO more directly, we summarize its end-to-end training process as Algorithm 1.

Algorithm 1 End-to-End Stochastic Predict-Then-Optimize

Require: Predicted distribution parameters $(\hat{\mu}, \hat{\sigma}^2)$, true cost vector c , learning rate η , number of Monte Carlo samples M

- 1: Initialize $\hat{\mu}$ and $\hat{\sigma}^2$ randomly
- 2: **while** training **do**
- 3: Predict the probability distribution parameters $\hat{\mu}_i, \hat{\sigma}_i^2$ using the model $f(\Theta)$
- 4: Sample $\{\epsilon_i \sim N(0, I)\}_{i=1}^M$
- 5: Compute Monte Carlo cost samples: $\hat{c}_i = \hat{\mu} + \sqrt{\hat{\sigma}^2} \epsilon_i$ for $i = 1, \dots, M$
- 6: **for** $i = 1, \dots, M$ **do**
- 7: Solve optimization: $u(2\hat{c}_i - c) \in \arg \min_{u \in \mathcal{U}} (2\hat{c}_i - c)^\top u$
- 8: **end for**
- 9: Compute subgradient of $\hat{\mu}$:

$$\partial_{\hat{\mu}} l_{SPTO+} \approx \frac{2}{M} \sum_{i=1}^M [u(c) - u(2\hat{c}_i - c)]$$

- 10: Compute subgradient of $\hat{\sigma}^2$:

$$\partial_{\hat{\sigma}^2} l_{SPTO+} \approx \frac{1}{M} \sum_{i=1}^M [u(c) - u(2\hat{c}_i - c)] \frac{\epsilon_i}{\sqrt{\hat{\sigma}^2}}$$

- 11: Update $\hat{\mu}$ and $\hat{\sigma}^2$:

$$\hat{\mu} \leftarrow \hat{\mu} - \eta \partial_{\hat{\mu}} l_{SPTO+}, \quad \hat{\sigma}^2 \leftarrow \hat{\sigma}^2 - \eta \partial_{\hat{\sigma}^2} l_{SPTO+}$$

- 12: Update the prediction model $f(\Theta)$

- 13: **end while**

TABLE I
PUMPS PARAMETERS

Station	Process	Pump Index	Renumber (j)	Flow (m ³ /h)	Power (kW)
Station A	Production	1# ~ 6#	1 ~ 3	4340	160
		7# ~ 8#	8 ~ 9	2088	75
	Distribution	1# ~ 3#	9 ~ 11	5500	1120
		4# ~ 6#	12 ~ 14	3100	630
Station B	Distribution	1# ~ 2#	15 ~ 16	1260	220
Station C	Distribution	1# ~ 2#	17 ~ 18	3500	450

V. EXPERIMENTS AND RESULTS DISCUSSION

A. Experimental Settings

Pump parameters settings: The parameters of all the pumps for the distribution and production process in Stations A ~ C in Fig. 1 are listed in table I.

Implementation Details: For solving the optimization problem, the hardware environment is a Windows personal computer with Intel(R) Core(TM) i5-13600k CPU @ 3.20 GHz, 32 GB RAM, and RTX 4070ti GPU. As for the model implementation, Pyomo is applied to construct the model, and the Gurobi solver is utilized to solve it. This section examines three main components: the effectiveness of the SPTO, the accuracy of the forecasting model, and the solution of the optimization model. Such an organization is to

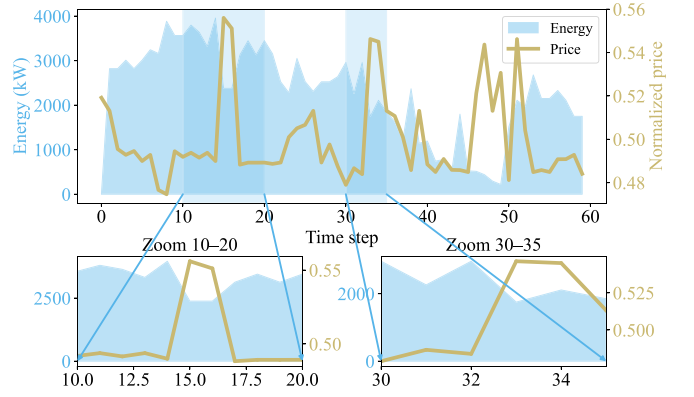


Fig. 3. Demonstration of energy usage and price trend with zoomed peak periods.

assess whether the proposed SPTO framework can reduce the optimality gap; whether the forecasting results are satisfactory, and whether the optimization problems are solved correctly as expected.

Baselines: To evaluate the performance of the proposed SPTO framework, we compare it with three baselines. Deterministic two-stage approach (DTS) [26] predicts uncertain parameters as fixed point values and solves the optimization problem based on these predictions. Stochastic two-stage approach (STS) [24] addresses the uncertainty using probabilistic forecasting models and optimizes the expected objective value. However, the distribution is not directly incorporated into the optimization stages. Smart predict-then-optimize (SPO) [21] integrates the optimization objective into the training process of the prediction model. Unlike DTS and STS, SPO aligns predictions with the optimization stages. However, it utilizes deterministic predictions and does not explicitly handle the uncertainty of parameter prediction errors.

B. Optimization Solution Analysis

Firstly, we investigate whether the proposed SPTO scheduling model is responsive to electricity prices by comparing energy consumption trends against dynamic price signals. Fig. 3 shows the total energy consumption and price over time, with two high-price intervals zoomed in. During these peak periods (steps 10–20 and 30–35), energy consumption drops significantly, indicating that the model strategically shifts operations away from periods of high expense. This aligns with the core design of the cost-aware SPTO framework.

Secondly, we validate the negative correlation between energy usage and price. As shown in Fig. 4, most sampled points cluster in the low-price region with high energy usage. A fitted regression line confirms the inverse trend, with a negative slope. This negative correlation supports the result in Fig. 3, showing that the scheduling model reduces load during expensive periods and concentrates operations when the price is low.

Thirdly, we examine the smoothness of the energy scheduling behavior. Fig. 5 presents the ramp rate curve and its histogram distribution. The curve shows that most fluctuations are mild and centered around zero. The inset histogram

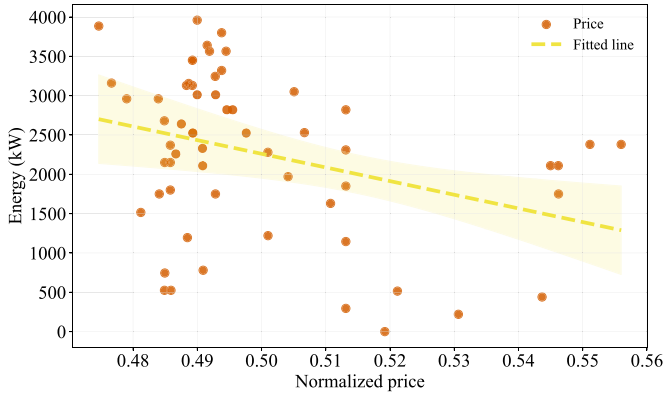


Fig. 4. Scatter comparison of energy usage and electricity price.

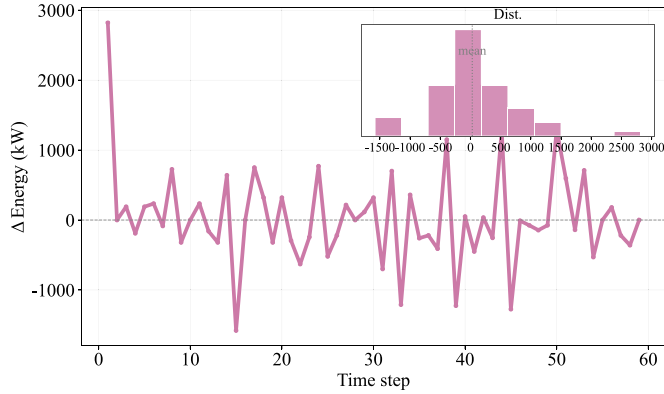


Fig. 5. Ramp rate distribution of energy consumption.

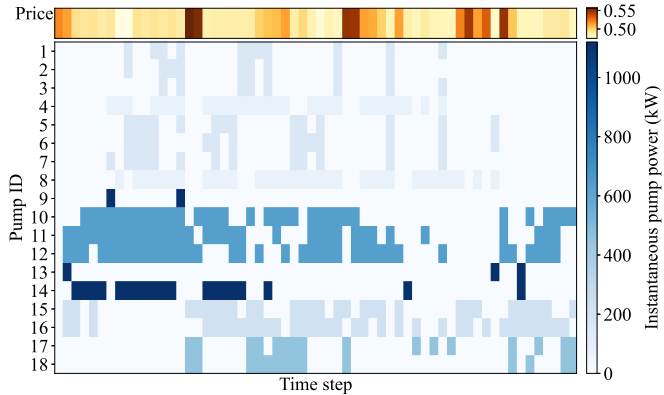
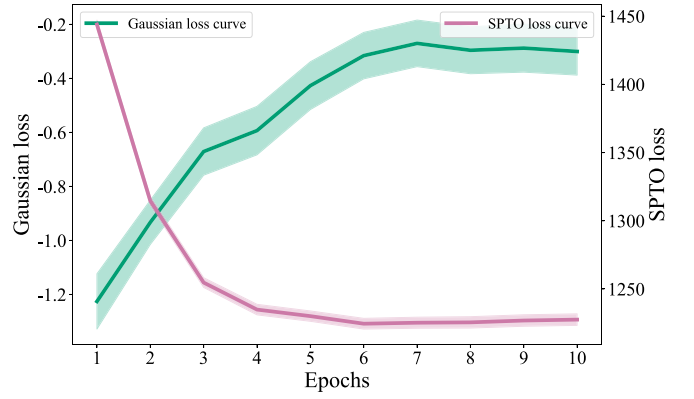


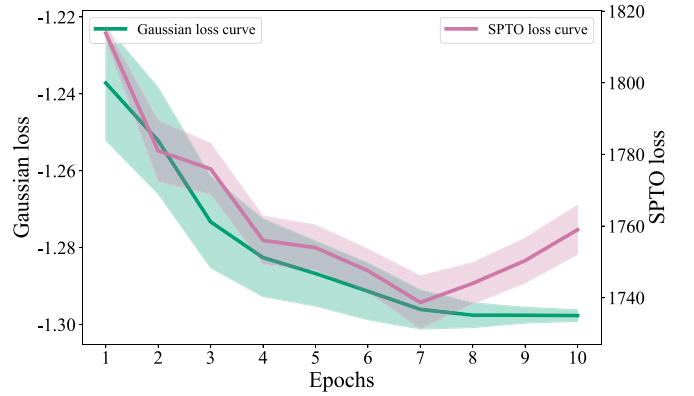
Fig. 6. Pump-level power usage heatmap and price overlap.

presents the distribution of ramp rates, with a mean close to zero and a narrow variance. This indicates that the model does not introduce many abrupt changes in energy usage. Smooth scheduling helps reduce operational stress on the equipment and avoids unexpected load shocks to the grid.

Also, we analyze the asynchronous coordination across different pump stations. Fig. 6 provides a heatmap of pump-level power usage with electricity price across time. Although not all pumps reduce activity uniformly during price peaks, the overall behavior reflects a coordinated yet flexible response strategy. This confirms the effectiveness of the decentralized scheduling policy under system-wide constraints.



(a) Training forecasting model with SPTO loss



(b) Training forecasting model with Gaussian loss

Fig. 7. Loss curve comparison while training with SPTO or Gaussian Loss.

Results demonstrated across Fig. 3 - Fig. 6 show that the SPTO framework reduces energy use during high-price periods and maintains smooth scheduling without sharp fluctuations. These results confirm that SPTO achieves cost-efficient and stable scheduling in real-world settings.

C. Effectiveness of SPTO Examination

Moreover, we would like to verify the misconception mentioned in the introduction that higher forecasting accuracy, as measured by the two-stage approach, does not necessarily suggest smaller decision regret. Fig. 7 presents a comparative analysis of training forecasting models using SPTO loss and Gaussian loss separately. In sub-figure (a), training with the SPTO loss (red line) leads to an increase in the Gaussian loss (blue line) of the prediction model, which increases from approximately -1.2 to -0.6 within 10 epochs. Meanwhile, sub-figure (b) shows that training with the Gaussian loss (blue line) ultimately leads to a higher SPTO loss (red line). The SPTO loss increases from 1740 to 1820.

These findings confirm that the SPTO loss effectively tunes the prediction and optimization stages. While Gaussian loss improves the prediction accuracy of traditional measurements, it fails to reduce decision regret. Instead, SPTO loss is able to reduce decision regret, emphasizing the importance of aligning prediction and optimization in an end-to-end approach.

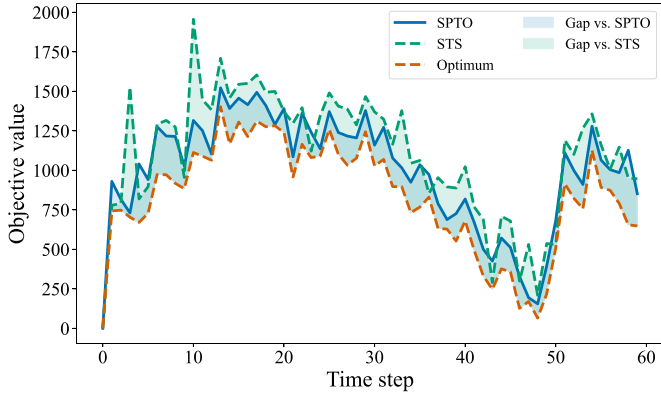


Fig. 8. Comparison of the SPTO and two-stage approaches.

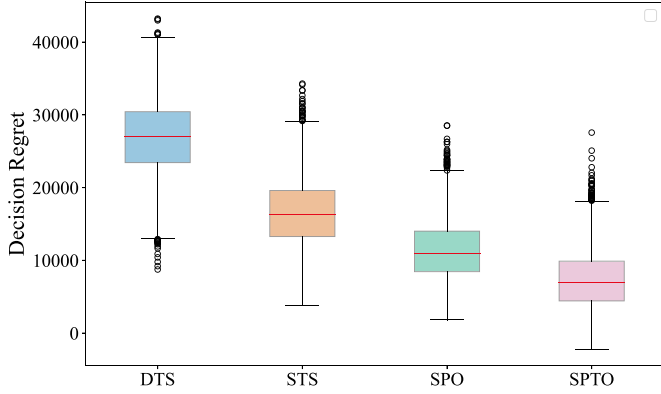


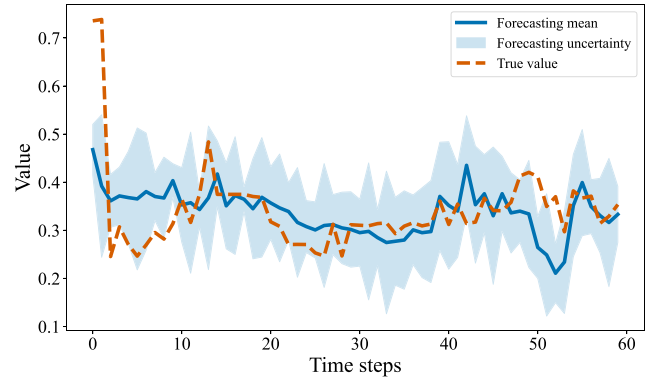
Fig. 9. Decision regret comparison of different approaches.

We also evaluate the effectiveness of the end-to-end SPTO method against the two-stage approach, investigating its application in stochastic optimization scenarios for minimizing the optimality gap. In Fig. 8, the x-axis represents the time step with 5 minutes for each step. The y-axis reflects the optimality gap of the different methods while optimizing on the time horizon.

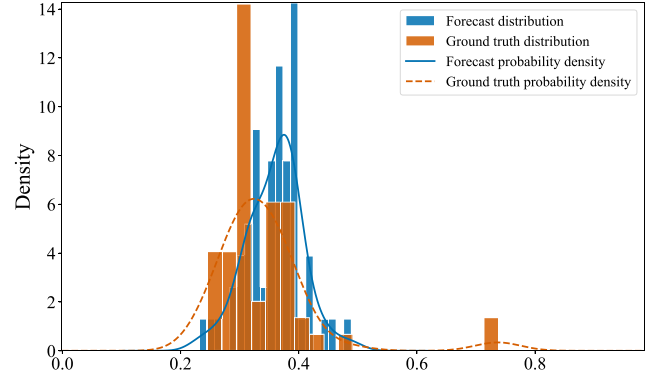
The results show that the SPTO framework (blue line) is very close to the optimum (red dashed line), with the optimal gap consistently remaining below 1000 units. In contrast, the two-stage approach (green dashed line) shows significantly higher gaps and greater variability. In addition, compared to the red shaded region, the narrower blue shaded region highlights the robustness of SPTO in minimizing the optimality gap. These findings suggest that the end-to-end SPTO approach provides solutions that are closer to the true optimum, which outperforms the two-stage approach.

To further evaluate the effectiveness of SPTO, we compare its decision regret against the deterministic two-stage approach (DTS), stochastic two-stage approach (STS), and SPO methods, as shown in Fig. 9. Each method is represented on the x-axis, while the y-axis reflects the decision regret.

The results show that SPTO has the lowest median decision regret and significantly less variability than the other methods. DTS and STS show higher medians and greater variability. SPO, on the other hand, although slightly better than the two-stage approach, still does not perform as well as SPTO.



(a) True vs forecasting with Uncertainty



(b) Probabilistic density of forecasting error

Fig. 10. Comparison with forecasting results and the true values.

The narrower interquartile range and fewer outliers for SPTO highlight its performance in minimizing decision regret. By directly taking into account the predictive distribution, SPTO demonstrates superiority in achieving smaller decision regret, making it a better choice for stochastic optimization problems than the traditional two-stage approach and deterministic SPO.

D. Evaluation of the Probabilistic Forecasting Model

Then, the performance of the applied probabilistic forecasting model is examined and shown in Fig. 10. The figure comprises two subplots that evaluate the performance and reliability of the probabilistic forecasting model. Subplot (a) compares the true values against the forecast mean, highlighting forecasting uncertainty over time. The green line represents the predicted mean, the purple dashed line shows the true values, and the shaded green area represents the forecasting uncertainty interval. Subplot (b) presents a visual representation of the probabilistic density of forecasting errors. The green bars represent the forecast distribution, the purple bars illustrate the ground truth distribution, and the dashed green line shows the probability density of the forecasting errors.

From the two subplots, we can visualize that the mean of the predicted values is similar to the true values, and the shape of the distribution is comparable. When considering the confidence interval, it can be observed that the true values mostly fall within the forecasting uncertainty range.

This implies that the probabilistic prediction model effectively estimates the uncertainty.

VI. CONCLUSION

This paper investigates the challenge of cost-efficient water-energy scheduling under electricity price uncertainty within the smart grid. To address the inherent misalignment between prediction and decision-making in traditional two-stage methods, we develop an end-to-end *Stochastic Predict-Then-Optimize* (SPTO) framework. By integrating a probabilistic forecasting model with a stochastic regret-minimization objective, the SPTO approach enables direct alignment between predictive accuracy and downstream operational decisions.

Unlike prior PTO studies limited to synthetic benchmarks, this work applies the SPTO framework to a real-world urban water-energy system, demonstrating its practical scalability and impact. The experimental results show that SPTO effectively mitigates cumulative decision errors induced by prediction uncertainty, leading to improved energy cost savings and load flexibility. Beyond optimizing pump operations, our findings also underscore the coupled dynamics between water infrastructure and power systems, advocating for integrated approaches to urban resource management in smart grid environments.

REFERENCES

- [1] T. Jiang, P. Ju, Z. Lin, Q. Wu, Z. Lin, and C. Y. Chung, "Competitive incentive mechanism for multi-agents in demand response via a hierarchical game considering joint uncertainties," *IEEE Trans. Smart Grid*, vol. 16, no. 4, pp. 2913–2925, Jul. 2025.
- [2] J. Xu, T. Zheng, Y. Dang, F. Yang, and D. Li, "Distributed deep reinforcement learning for data-driven water heater model in smart grid," *IEEE Trans. Smart Grid*, vol. 16, no. 4, pp. 2900–2912, Jul. 2025.
- [3] Q. Shi, Y. Yan, W. Liu, B. Zeng, Z. Wang, and F. Li, "Resilience assessment of urban power network considering multi-stage electrical-water fault propagation," *IEEE Trans. Power Syst.*, vol. 40, no. 4, pp. 2819–2831, Jul. 2025.
- [4] A. Stuhlmacher, S. Guikema, and J. L. Mathieu, "Assessing power and water network resilience when water pumps provide frequency regulation," *IEEE Trans. Power Syst.*, vol. 40, no. 5, pp. 3833–3845, Sep. 2025.
- [5] S. Wu, J. Wang, H. Xu, S. Zhao, and J. Xu, "Criticoder: An end-to-end uncertain regression network for robust macroscopic pressure models in water distribution systems," *IEEE Trans. Comput. Social Syst.*, vol. 11, no. 2, pp. 2222–2233, Apr. 2024.
- [6] A. Stuhlmacher and J. L. Mathieu, "Chance-constrained water pumping to manage water and power demand uncertainty in distribution networks," *Proc. IEEE*, vol. 108, pp. 1640–1655, Sep. 2020.
- [7] J. Bhardwaj, J. P. Krishnan, D. F. Marin, B. Beferull-Lozano, L. R. Cenkeramaddi, and C. Harman, "Cyber-physical systems for smart water networks: A review," *IEEE Sensors J.*, vol. 21, pp. 26447–26469, Dec. 2021.
- [8] W. Zhao, T. H. Beach, and Y. Rezgui, "Optimization of potable water distribution and wastewater collection networks: A systematic review and future research directions," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 46, no. 5, pp. 659–681, May 2016.
- [9] Y. Cao, B. Zhou, C. Y. Chung, Z. Shuai, Z. Hua, and Y. Sun, "Dynamic modelling and mutual coordination of electricity and watershed networks for spatio-temporal operational flexibility enhancement under rainy climates," *IEEE Trans. Smart Grid*, Vol. 14, no. 5, pp. 3450–3464, Sep. 2023.
- [10] R. Koh, J. Kern, and S. Galelli, "Hard-coupling water and power system models increases the complementarity of renewable energy sources," *Appl. Energy*, vol. 321, Sep. 2022, Art. no. 119386.
- [11] A. B. Bianchi, J. Willet, H. Rijnaarts, and S. S. Torbaghan, "Distribution robust water-based demand side management in power transmission networks," *Sustain. Energy, Grids Netw.*, vol. 36, Oct. 2023, Art. no. 101232.
- [12] B. Tasseff et al., "Polyhedral relaxations for optimal pump scheduling of potable water distribution networks," *INFORMS J. Comput.*, vol. 36, no. 4, pp. 1040–1063, 2024.
- [13] E. A. M. Cesena et al., "Quantifying the impacts of modelling assumptions on accuracy and computational efficiency for integrated water-energy system simulations under uncertain climate," *IEEE Trans. Sustain. Energy*, vol. 13, no. 3, pp. 1370–1382, Jul. 2022.
- [14] Y. Yao, C. Li, H. M. Tai, and C. Shao, "Modeling of water transfer delay for optimal operation of integrated electrical-water system," *IEEE Trans. Power Syst.*, vol. 38, no. 1, pp. 331–348, Jan. 2023.
- [15] K. Oikonomou and M. Parvania, "Optimal coordinated operation of interdependent power and water distribution systems," *IEEE Trans. Smart Grid*, vol. 11, no. 6, pp. 4784–4794, Nov. 2020.
- [16] Y. C. Hung and G. Michailidis, "Modeling and optimization of time-of-use electricity pricing systems," *IEEE Trans. Smart Grid*, vol. 10, no. 4, pp. 4116–4127, Jul. 2019.
- [17] C. Gu, Y. Pan, R. Liu, and Y. Chen, "Learning and optimization for price-based demand response of electric vehicle charging," 2024, *arXiv:2404.10311v1*.
- [18] S. Zuloaga and V. Vittal, "Integrated electric power/water distribution system modeling and control under extreme mega drought scenarios," *IEEE Trans. Power Syst.*, vol. 36, no. 1, pp. 474–484, Jan. 2021.
- [19] A. Miraki, P. Parviainen, and R. Arghandeh, "Electricity demand forecasting at distribution and household levels using explainable causal graph neural network," *Energy AI*, vol. 16, May 2024, Art. no. 100368. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S266654682400034X>
- [20] J. Mandi, T. Guns, E. Demirović, and P. J. Stuckey, "Smart predict-and-optimize for hard combinatorial optimization problems," in *Proc. 34th AAAI Conf. Artif. Intell.*, vol. 34, 2020, pp. 1603–1610.
- [21] A. N. Elmachtoub and P. Grigas, "Smart 'predict, then Optimize,'" *Manage. Sci.*, vol. 68, no. 1, pp. 9–26, 2022.
- [22] B. Tang and E. B. Khalil, "PyEPO: A PyTorch-based end-to-end predict-then-optimize library for linear and integer programming," Jun. 2022, *arXiv:2206.14234*.
- [23] H. Geng et al., "Rethinking and benchmarking predict-then-optimize paradigm for combinatorial optimization problems," Nov. 2023, *arXiv:2311.07633*.
- [24] P. Donti, B. Amos, and J. Z. Kolter, "Task-based end-to-end model learning in stochastic optimization," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–15.
- [25] B. Amos and J. Z. Kolter, "OptNet: Differentiable optimization as a layer in neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 136–145.
- [26] J. Mandi and T. Guns, "Interior point solving for LP-based prediction+Optimisation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 7272–7282.
- [27] A. M. Ferber et al., "Surco: Learning linear surrogates for combinatorial nonlinear optimization problems," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 10034–10052.
- [28] J. Zhang, E. Shan, L. Wu, J. Yin, L. Yang, and Z. Gao, "An end-to-end predict-then-optimize clustering method for stochastic assignment problems," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 9, pp. 12605–12620, Sep. 2024.
- [29] W. Guo et al., "Integrating end-to-end prediction-with-optimization for distributed hydrogen energy system scheduling," in *Proc. IEEE 20th Int. Conf. Autom. Sci. Eng. (CASE)*, 2024, pp. 2774–2779.
- [30] D. Salinas, V. Flunkert, J. Gasthaus, and T. Januschowski, "DeepAR: Probabilistic forecasting with autoregressive recurrent networks," *Int. J. Forecast.*, vol. 36, no. 3, pp. 1181–1191, 2020.
- [31] G. Memarzadeh and F. Keynia, "Short-term electricity load and price forecasting by a new optimal LSTM-NN based prediction algorithm," *Elect. Power Syst. Res.*, vol. 192, Mar. 2021, Art. no. 106995.
- [32] S. Wu, J. Wang, H. Xu, S. Zhao, and J. Xu, "Knowledge-based bi-correction model for achieving effective lag-free characteristic on daily urban water demand forecasting," *Expert Syst. Appl.*, vol. 255, Dec. 2024, Art. no. 124508. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0957417424013757>
- [33] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–15.